2015-2016 – Econ590
Topics in Macroeconomics

Lecture 1 : Business Cycle Models : The Current Consensus (Part B)

Paul Beaudry & Franck Portier
franck.portier@TSE-fr.eu

Vancouver School of Economics

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Changes from version 1.0 are in red
Changes from version 1.1 are in purple

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5. The New Keynesian Model

- We start by quickly reviewing facts about money and economic activity
- Then we look at a monetary model without or with nominal rigidities

GEORGE MCCANDLESS *and* WARREN WEBER, *"Some Monetary Facts", Minneapolis Fed QR, 1995*

- McCandless and Weber : 110 countries over 30 years
- Compute the long-run geometric average rate of growth for :
  - the standard measure of production : gross domestic product adjusted for inflation (real GDP) ;
  - a standard measure of the general price level : consumer prices ;
  - three commonly used definitions of money (M0, M1, and M2)
- They also look at 2 more homogenous sub samples : 21 OECD countries and 14 Latin American countries.
- Three main results :

5.1.1. Long run facts

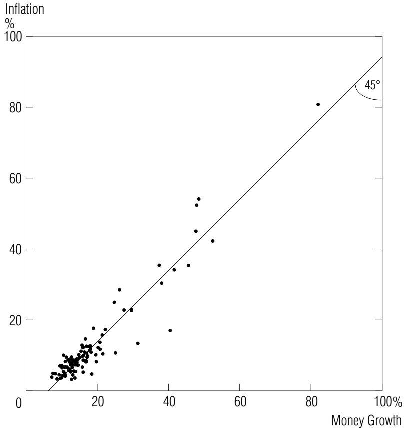
Result 1 – Money Growth and Inflation : *In the long run, there is a high (almost unity) correlation between the rate of growth of the money supply and the rate of inflation. This holds across three definitions of money and across the full sample of countries and two subsamples.*

5.1.1. Long run facts

Chart 1

Money Growth and Inflation:
A High, Positive Correlation

Average Annual Rates of Growth in M2 and in Consumer Prices
During 1960–90 in 110 Countries



Source: International Monetary Fund

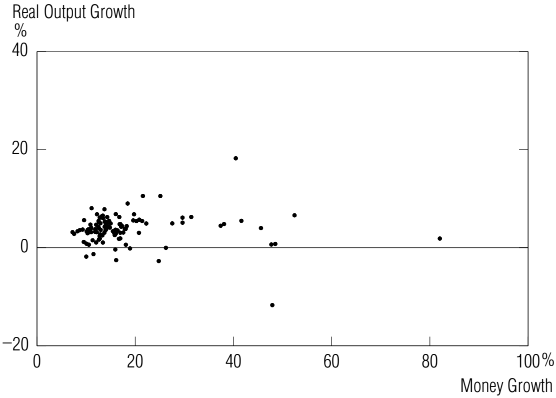
5.1.1. Long run facts

Result 2 –Money Growth and Real Output Growth : *In the long run, there is no correlation between the growth rates of money and real output. This holds across all definitions of money, but not for a subsample of OECD countries, where the correlation seems to be positive.*

Chart 2

Money and Real Output Growth: No Correlation in the Full Sample . . .

Average Annual Rates of Growth in M2
and in Nominal Gross Domestic Product, Deflated by Consumer Prices
During 1960–90 in 110 Countries

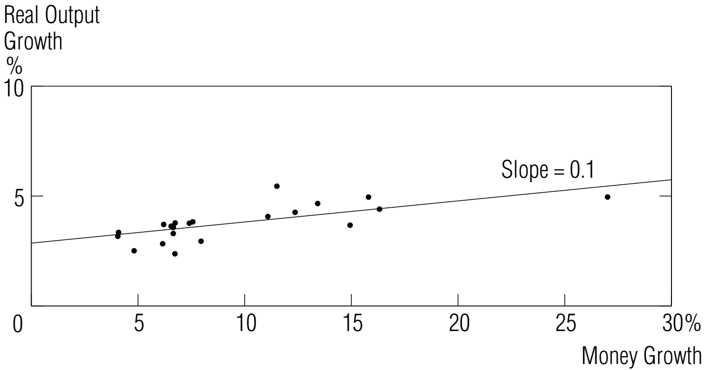


Source: International Monetary Fund

Chart 3

. . . But a Positive Correlation in the OECD Subsample

Average Annual Rates of Growth in M0
and in Nominal Gross Domestic Product, Deflated by Consumer Prices
During 1960–90 in 21 Countries



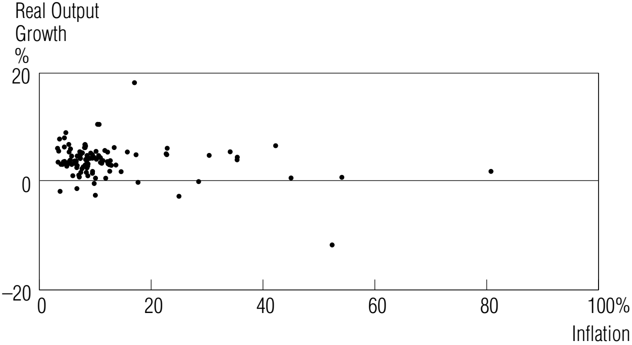
Source: International Monetary Fund

Result 3 –Inflation and Real Output Growth : *In the long run, there is no correlation between inflation and real output growth. This finding holds across the full sample and both subsamples.*

Chart 4

Inflation and Real Output Growth: No Correlation

Average Annual Rates of Growth in Consumer Prices
and in Nominal Gross Domestic Product, Deflated by Consumer Prices
During 1960–90 in 110 Countries



Source: International Monetary Fund

5.1.2. Short Run Evidence

- It is much more difficult to get non controversial results in the short run, because money is partly endogenous.
- Money and output may vary because money causes output, but also because output causes money (banking system), or as a common response to a third variable (technology)

5.1.2. Short Run Evidence

Traditional approach

- Unconditional correlations suggest that the correlation is positive and that money is leading
- The correlation is larger for broader aggregate like M2, which is more endogenous.
- Seminal work of Friedman and Schwartz (1963) : money matters for BC because it leads output
- But Tobin (1970) : *post hoc ergo propter hoc* (reverse causality) $\rightsquigarrow$ need for conditional correlations.
- On top of that, monetary aggregates need not to be always the instrument of the monetary authorities (money market interest rate is now the most commonly used instrument)

- SIMS (1972) : Does money Granger-cause output :

$$y_t = y_0 + \sum_{i>0} a_i m_{t-i} + \sum_{i>0} b_i y_{t-i} + \sum_{i>0} c_i z_{t-i} + e_t$$

- If all a_i are zeros, then money does not cause output. $\rightsquigarrow$ evidence are that it does, but results depends on the specification (lags, extra variables z, detrending, etc...)
- Extra work by BARRO (1978) : does both anticipated and unanticipated parts of money matter fore real output $\rightsquigarrow$ only the unanticipated one for BARRO (but some contradictory findings later)

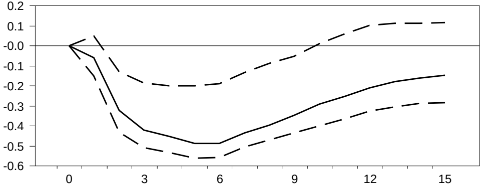
- The idea here is to "purge" the monetary policy instrument from responses to other shocks, so that the "pure" response to a monetary shocks can be estimated.
- We will spend much more time on this later

- The following figures display the response of various variables to a one standard deviation monetary shock ϵ^{s}.
- The x-axis represents quarters after the shock (the shock is in quarter 0)
- The y-axis is in percentage deviation from the trajectory without the shock
- "MP" means Monetary Policy shock. It is a contractionary shock.

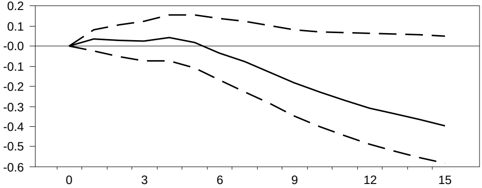
5.1.2. Short Run Evidence

Monetary VAR's

MP Shock => Y



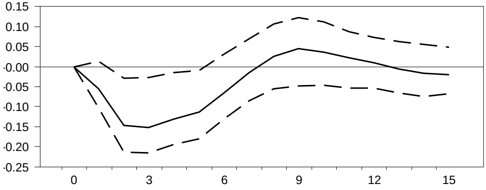
MP Shock => Price



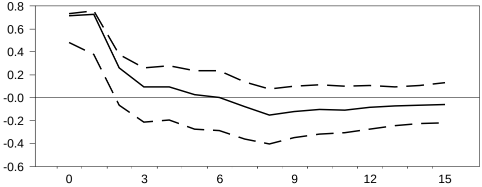
5.1.2. Short Run Evidence

Monetary VAR's

MP Shock => Pcom



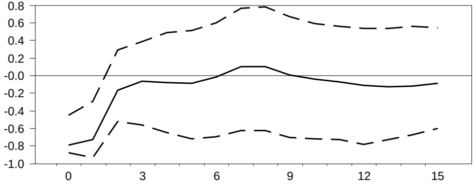
MP Shock => FF



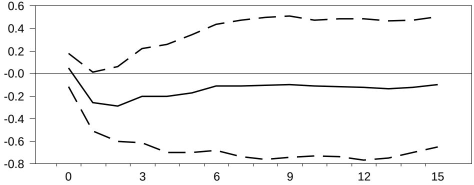
5.1.2. Short Run Evidence

Monetary VAR's

MP Shock => NBR

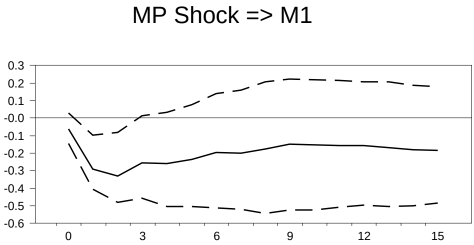


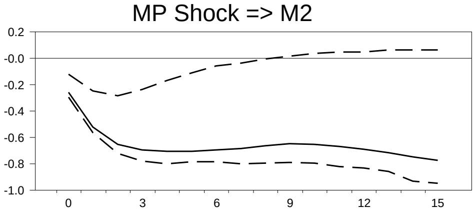
MP Shock => TR



5.1.2. Short Run Evidence

Monetary VAR's



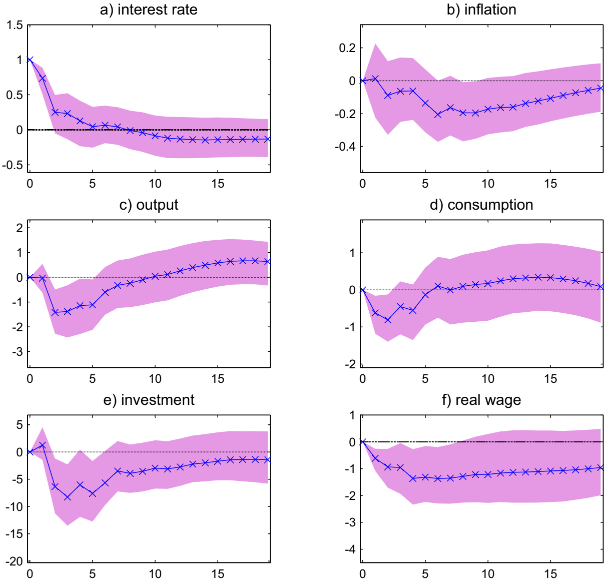


- Results : Not much response of prices in the short run, expansionary effect, liquidity effect $\rightsquigarrow$ short run non neutrality
- One can extend CEE work by adding more variables in the VAR

- The following figures display the response of various variables to a one standard deviation monetary shock ϵ^s.
- The x-axis represents quarters after the shock (the shock is in quarter 0)
- The y-axis is in percentage deviation from the trajectory without the shock
- Those estimates on US.data are proposed by André MEIER and Gernot J. MÜLLER in a ECB working paper (number 500, 2005)

5.1.2. Short Run Evidence

Monetary VAR's



- Objective :

$$E_0 \sum_{t=0}^{\infty} \beta^t U\left(C_t, N_t, \frac{M_t}{P_t}\right)$$

- Budget Constraint :

$$P_t C_t + M_t + Q_t B_t \leq B_{t-1} + W_t N_t + M_{t-1} + T_t$$

- T_t gathers lump sum transfers (money, taxes) and dividends
- B_t is a riskless one-period bond that pays 1 unit of money in $t+1$
- $\mathcal{A}_t = B_{t-1} + M_{t-1}$, $\lim_{T\to\infty} E_t \mathcal{A}_T \geq 0$

- FOC :

$$\begin{cases} -\frac{U_{n,t}}{U_{c,t}} & = & \frac{W_t}{P_t} \\ Q_t & = & \beta E_t\left[\frac{U_{c,t+1}}{U_{c,t}}\frac{P_t}{P_{t+1}}\right] \\ \frac{U_{m,t}}{U_{c,t}} & = & 1-Q_t \end{cases}$$

- We assume in the following : $U = \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\phi}}{1+\phi} + \frac{\left(\frac{M_t}{P_t}\right)^{1-\nu}}{1-\nu}$
- Then the FOC become

$$\begin{cases} \frac{W_t}{P_t} & = & C_t^{\sigma} N_t^{\phi} & (a) \\ \frac{M_t}{P_t} & = & C_t^{\sigma/\nu}(1-Q_t)^{-1/\nu} & (b) \\ 1 & = & \beta E_t\left[\frac{1}{Q_t}\left(\frac{C_{t+1}}{C_t}\right)^{-\sigma}\frac{P_t}{P_{t+1}}\right] & (c) \end{cases}$$

- We log-linearize those equations (some are already loglinear, we just need to take logs).
- We use the notation $x = \log X$
- (a) gives $w_t - p_t = \sigma c_t + \phi n_t$

- Define $i_t = \log \frac{1}{Q_t} = -\log Q_t$: nominal interest rate ($Q_t = \exp(-i_t))$)
- Define $\pi_{t+1} = p_{t+1} - p_t = \log\left(\frac{P_{t+1}}{P_t}\right)$: inflation
- Let $\rho = \log \frac{1}{\beta}$
- Take (c) and replace X by $\exp(\log(X)) = \exp(x)$

5.3.1. Households

Linearization of Equation (c)

- Equation (c) :

$$1 = \beta E_t \left[\frac{1}{Q_t} \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} \right]$$

$$\iff$$

$$1 = E_t \left[\exp \left\{ \log \left(\frac{1}{Q_t} \right) - \sigma \log \left(\frac{C_{t+1}}{C_t} \right) - \log \left(\frac{P_{t+1}}{P_t} \right) + \log(\beta) \right\} \right]$$

$$\iff$$

$$1 = E_t \left[\exp \left\{ i_t - \sigma \Delta c_{t+1} - \pi_{t+1} - \rho \right\} \right]$$

- Consider the non stochastic perfect foresight steady state with constant inflation π and constant real growth γ.
- (c) implies $1 = \exp(i - \sigma\gamma - \pi - \rho)$
- and therefore $i - \sigma\gamma - \pi - \rho = 0$

5.3.1. Households

How to take a first order expansion of $\exp(x)$ around 0 ?

- Taylor expansion :

$$\exp(x_t) \approx \exp(0) + \exp(0) \times (x_t - 0) = 1 + x_t$$

- Here :

$$\exp(i_t - \sigma \Delta c_{t+1} - \pi_{t+1} - \rho) \approx 1 + i_t - \sigma \Delta c_{t+1} - \pi_{t+1} - \rho$$

- and then

$$1 = E_t \left[\exp\left\{i_t - \sigma \Delta c_{t+1} - \pi_{t+1} - \rho\right\}\right]$$

$$\iff c_t = E_t\left[c_{t+1}\right] - \frac{1}{\sigma}(\underbrace{i_t - E_t\left[\pi_{t+1}\right]}_{r_t} - \rho)$$

- Take (b) :

$$\frac{M_t}{P_t} = C_t^{\sigma/\nu}(1 - Q_t)^{-1/\nu} \qquad (b)$$

$$\Longleftrightarrow \exp(m_t - p_t) = \exp\left(\frac{\sigma}{\nu}c_t - \frac{1}{\nu}\log(1 - \exp(-i_t))\right)$$

- and therefore

$$m_t - p_t = \frac{\sigma}{\nu}c_t - \frac{1}{\nu}\log(1 - \exp(-i_t))$$

- which is also $\quad m_t - p_t = \frac{\sigma}{\nu}c_t - \eta i_t$ with $\eta = \frac{1}{\nu(\exp(i) - 1)} \approx \frac{1}{\nu i}$

- We assume in the following a unit income (consumption) elasticity of money demand : $\sigma = \nu$
- The optimal Hh behavior is then summarized by the three following equations (+ the BC)

$$\begin{cases} \text{labor supply :} & & \\ w_t - p_t & = & \sigma c_t + \phi n_t & (1) \\ \text{money demand :} & & \\ m_t - p_t & = & c_t - \eta i_t & (2) \\ \text{cons./saving :} & & \\ c_t & = & E_t\left[c_{t+1}\right] - \frac{1}{\sigma}(i_t - E_t\left[\pi_{t+1}\right] - \rho) & (3) \end{cases}$$

5.3.2. Firms

- $Y_t = A_t N_t^{1-\alpha}$
- $\Pi_t = P_t Y_t - W_t N_t$
- FOC : $\frac{W_t}{P_t} = (1-\alpha) A_t N_t^{-\alpha}$
- From which we obtain the log-linear labor demand (+ the production function) :

$$\begin{cases} \text{labor demand :} & \\ w_t - p_t = a_t - \alpha n_t + \log(1-\alpha) & (4) \\ \text{production function :} & \\ y_t = a_t + (1-\alpha) n_t & (5) \end{cases}$$

5.3.3. Equilibrium – Real Variables

- The real equilibrium is given by the good market equilibrium and the labor market one (+ bonds market clearing that implies $B_t = 0 \ \forall t$).

$$
\begin{cases}
\text{labor market :} & & & & \\
a_t - \alpha n_t + \log(1-\alpha) & = & w_t - p_t & = & \sigma c_t + \phi n_t \\
\text{good market :} & & & & \\
c_t & = & y_t & = & a_t + (1-\alpha) n_t
\end{cases}
$$

5.3.3. Equilibrium – Real Variables

- which gives

$$\begin{cases} n_t &= \psi_{na} a_t + \theta_n \\ y_t &= \psi_{ya} a_t + \theta_y \end{cases}$$

with $\psi_{na} = \frac{1-\sigma}{\sigma(1-\alpha)+\phi+\alpha}$, $\psi_{ya} = \frac{1+\phi}{\sigma(1-\alpha)+\phi+\alpha}$, $\theta_n = \frac{\log(1-\alpha)}{\sigma(1-\alpha)+\phi+\alpha}$, $\theta_y = (1-\alpha)\theta_n$.

- and then

$$\begin{cases} r_t &= i_t - E_t\left[\pi_{t+1}\right] \\ &= \sigma E_t\left[\Delta y_{t+1}\right] + \rho \\ &= \rho + \sigma \psi_{ya} E_t\left[\Delta a_{t+1}\right] \\ & \\ \omega_t &= w_t - p_t \\ &= a_t - \alpha n_t + \log(1-\alpha) \\ &= \psi_{\omega a} a_t + \theta_\omega \end{cases}$$

with $\psi_{a\omega} = \frac{\sigma+\phi}{\sigma(1-\alpha)+\phi+\alpha}$, $\theta_\omega = \frac{(\sigma(1-\alpha)+\phi)\log(1-\alpha)}{\sigma(1-\alpha)+\phi+\alpha}$.

- Equilibrium real allocations are independent of monetary policy
- A tech. shock increases y,
- A tech. shock has an ambiguous effect on n (wealth versus substitution effect)
- A tech. shock has an ambiguous effect on r , depending on whether $E_t a_{t+1} \gtrless a_t$

5.3.4. Equilibrium – Nominal Variables

- We have in equilibrium the "Fisherian" (Irving Fisher) equation : $i_t = E_t \pi_{t+1} + r_t$
- Monetary policy is about :
- choosing i (and therefore choosing π, as r is given from the real side of the economy)
- **or** choosing m, which determines p, π and i (using money demand $m_t - p_t = \frac{\sigma}{\nu} y_t - \eta i_t$.

5.3.4. Equilibrium – Nominal Variables

A Nominal Interest Rate Policy

- The "Monetary Authorities" control i, that follows an arbitrary exogenous stationary process $\{i_t\}$
- w.l.o.g, $\{i_t\}$ has mean ρ, and $\gamma = 0$, which is consistent with zero-inflation SS.
- A particular case is $i_t = i = \rho$
- With given $\{i_t\}$, the Fisher equation implies

$$E_t \pi_{t+1} = \underbrace{i_t}_{\text{determined by monetary policy}} - \underbrace{r_t}_{\text{determined from the real side of the model}} \tag{6}$$

- The only equation that determines money is actually pinning down expected expectation, but not actual inflation

- Let ξ_t be any *non fundamental* shocks (sunspot) s.t. $E_t\xi_{t+1} = 0$,
- Then $p_{t+1} = p_t + i_t - r_t + \xi_{t+1}$
- The price level is *indeterminate*, as it can be affected by any non-fundamental shock.
- Indeterminacy is only nominal
- The money supply that implements the interest rate policy can be recovered from money demand (+ the fact that money demand = money supply in equilibrium) :

$$m_t = p_t + y_t - \eta i_t$$

- Assume $i_t = \rho + \phi_\pi \pi_t$, $\qquad \phi_\pi \geq 0$
- Using (6),

$$\phi_\pi \pi_t = E_t \pi_{t+1} + \underbrace{\widehat{r}_t}_{r_t - \rho}$$

- The solution of this equation depends on the value of ϕ_π

- If $\phi_\pi > 1$:
  - In that case, we iterate forward :

$$\pi_t = \sum_{j=0}^{\infty} \phi_\pi^{-(j+1)} E_t \widehat{r}_{t+j}$$

    which fully determines π_t.
  - If for example

$$a_t = \rho_a a_{t-1} + \varepsilon_t^a,$$

    then

$$\widehat{r}_t = -\sigma\psi_{ya}(1-\rho_a)a_t$$

    and

$$\pi_t = \frac{\sigma\psi_{ya}(1-\rho_a)}{\phi_\pi - \rho_a} a_t.$$

  - The larger is ϕ_π, the smaller the volatility of inflation

- If $\phi_\pi < 1$:
  - In that case,

$$E_t \pi_{t+1} = \phi_\pi \pi_t - \widehat{r}_t.$$

  - Again, any non fundamental shocks ξ_t s.t. $E_t \xi_{t+1} = 0$ can be added to inflation :

$$\pi_{t+1} = \phi_\pi \pi_t - \widehat{r}_t + \xi_{t+1},$$

and inflation and the price level are indeterminate.

- We can derive a result that holds in a wide range of environnements :
- TAYLOR principle : *The Monetary Authorities must respond "aggressively" to inflation ($\phi_\pi > 1$) to guarantee determinacy of inflation.*

5.3.5. Monetary Rule

- Consider that the Monetary Authorities directly choose a path $\{m_t\}$
- From the money demand equation

$$m_t - p_t = y_t - \eta i_t,$$

we get

$$i_t = \frac{1}{\eta}(m_t - p_t - y_t). \tag{7}$$

- (6) writes

$$i_t = E_t p_{t+1} - p_t + r_t$$

- Combining with (7) :

$$\frac{1}{\eta}(m_t - p_t - y_t) = E_t p_{t+1} - p_t + r_t$$

$$\iff p_t = \frac{\eta}{1+\eta} E_t p_{t+1} + \frac{1}{1+\eta} m_t + u_t$$

where $u_t = (1+\eta)^{-1}(\eta r_t - y_t)$ is independent from m_t.

5.3.5. Monetary Rule

- Assuming $\eta > 0$, then $\frac{\eta}{1+\eta} < 1$, we can solve forward to get

$$p_t = \frac{1}{1+\eta} \sum_{j=0}^{\infty} \left(\frac{\eta}{1+\eta} \right)^j E_t m_{t+j} + \widehat{u}_t$$

with $\widehat{u}_t = \sum_{j=0}^{\infty} \left(\frac{\eta}{1+\eta} \right)^j E_t u_{t+j}$.

- The price level is fully determined, and can be written as

$$p_t = m_t + \sum_{j=1}^{\infty} \left(\frac{\eta}{1+\eta} \right)^j E_t \Delta m_{t+j} + \widehat{u}_t$$

- Using the money demand equation, we can obtain the solution for the nominal interest rate :

$$\begin{aligned} i_t &= \eta^{-1}(y_t - (m_t - p_t)) \\ &= \eta^{-1} \sum_{j=1}^{\infty} \left(\frac{\eta}{1+\eta}\right)^j E_t \Delta m_{t+j} + \widetilde{u}_t \end{aligned}$$

with $\widetilde{u}_t = \eta^{-1}(\widehat{u}_t + y_t)$.

5.3.5. Monetary Rule

Example

- Assume $\Delta m_t = \rho_m \Delta m_{t-1} + \varepsilon_t^m$ with $0 < \rho_m < 1$
- Then $p_t = m_t + \frac{\eta\rho_m}{1+\eta(1-\rho_m}\Delta m_t$
- Note that
$$\frac{\partial p_t}{\partial \varepsilon_t^m} = 1 + \frac{\eta\rho_m}{1+\eta(1-\rho_m} > 1$$
- The price level is predicted to respond more than one to one wrt money shocks $\rightsquigarrow$ not supported by the data, that show price stickiness.

5.3.6. Optimal Monetary Policy

- Money has no impact on real allocations...
- but $\frac{M}{P}$ enters the utility function.
- Let's solve a Social Planner problem.
- Note that the SP problem is static

$$\begin{array}{ll} \max_{C,M,N} & U\left(C_t, N_t, \frac{M_t}{P_t}\right) \\ \text{s.t.} & C_t = A_t N_t^{1-\alpha} \end{array}$$

- FOC :

$$\begin{cases} -\frac{U_{n,t}}{U_{c,t}} = (1-\alpha) A_t N_t^{-\alpha} & (aa) \\ U_{m,t} = 0 & (bb) \end{cases}$$

- (aa) holds in competitive equilibrium
- (bb) means that, given that producing real balances is free, the SP should satiate the agents ("choose $P = 0$")
- the competitive equivalent of (bb) is

$$\frac{U_{m,t}}{U_{c,t}} = 1 - Q_t = 1 - \exp(i_t)$$

- To implement a social optimum, monetary policy should set $i_t = 0 \ \forall t$, so that $\pi_t = -\rho$.

- In a wide variety of environments, we obtain the following rule :
- FRIEDMAN rule : *Deflation is optimal, at a rate equal to minus the real interest rate. This guarantees that the nominal interest rate (which is the opportunity cost of holding money) is zero.*

- A rule $i_t = 0$ leads to indeterminate inflation and prices.
- The rule $i_t = \phi(r_{t-1} + \pi_t)$ with $\phi > 1$ allows for determinacy, and $i_t = r_t + E_t \pi_{t+1}$ implies

$$E_t i_{t+1} = \phi E_t[r_t + \pi_{t+1}] = \phi i_t \Rightarrow i_t = 0 \quad \text{(solving forward)}$$

and inflation is determined as $\pi_t = -r_{t-1}$

- Objective :

$$E_0 \sum_{t=0}^{\infty} \beta^t V\left(C_t, N_t, \frac{M_t}{P_t}\right)$$

- We assume in the following : $V = \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1-\phi}}{1-\phi} + \chi \frac{\left(\frac{M_t}{P_t}\right)^{1-\sigma}}{1-\sigma}$
- We also assume $\chi \to 0$ so that $V\left(C_t, N_t, \frac{M_t}{P_t}\right) \approx U\left(C_t, N_t\right)$ but there exist a well-defined money demand function.
- Budget Constraint :

$$P_t C_t + M_t + Q_t B_t \leq B_{t-1} + W_t N_t + M_{t-1} + T_t$$

- $\mathcal{A}_t = B_{t-1} + M_{t-1}$, $\lim_{T\to\infty} E_t \mathcal{A}_T \geq 0$

- FOC :

$$
\begin{cases}
\text{labor supply :} & & \\
w_t - p_t & = & \sigma c_t + \phi n_t \qquad (1) \\
\text{money demand :} & & \\
m_t - p_t & = & c_t - \eta i_t \qquad (2) \\
\text{cons./saving :} & & \\
c_t & = & E_t\left[c_{t+1}\right] - \frac{1}{\sigma}(i_t - E_t\left[\pi_{t+1}\right] - \rho) \qquad (3)
\end{cases}
$$

- We assume that C is a basket of a continuum of consumption goods which are imperfect substitutes ($0 < \varepsilon < 1$, elaticitity of substitution $= \frac{1}{\varepsilon}$) :

$$C_t = \left(\int_0^1 C_t(i)^{1-\frac{1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon - 1}}$$

with prices $P_t(i)$ such that $P_t C_t = \int_0^1 P_t(i) C_t(i) di$

- Optimal composition of the basket is obtained by following the following program :

$$\begin{array}{ll} \min & \int_0^1 P_t(i) C_t(i) di \\ \text{s.t.} & \left(\int_0^1 C_t(i)^{1-\frac{1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon - 1}} \geq C_t \end{array}$$

- The solution of this problem is

$$C_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\varepsilon} C_t \tag{e}$$

with

$$P_t = \left(\int_0^1 P_t(i)^{1-\varepsilon} \right)^{\frac{1}{1-\varepsilon}}$$

- I am dropping the time subscript

$$\min \int_0^1 P_i C_i di \qquad s.t. \quad \left(\int_0^1 (C_i)^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}} \geq C \qquad (\lambda)$$

- The FOC of this program with respect to C_j gives

$$P_j = \lambda C_j^{-\frac{1}{\varepsilon}} \left(\int_0^1 (C_i)^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}-1}$$

- Taking the ratio of the FOC with respect to i with the FOC with respect to j gives

$$\frac{P_i}{P_j} = \left(\frac{C_i}{C_j} \right)^{-\frac{1}{\varepsilon}}$$

which is equivalent to

$$C_i = C_j \left(\frac{P_i}{P_j} \right)^{-\varepsilon}$$

5.4.1 Household

Optimal Composition of C–Details of the algebra

- Use this expression of C_i in the constraint (that is binding)

$$\left(\int_0^1 (C_i)^{\frac{\varepsilon-1}{\varepsilon}} di\right)^{\frac{\varepsilon}{\varepsilon-1}} = C,$$

  to obtain

$$C = \left[\int_0^1 \left(C_j \left(\frac{P_i}{P_j}\right)^{-\varepsilon}\right)^{\frac{\varepsilon-1}{\varepsilon}} di\right]^{\frac{\varepsilon}{\varepsilon-1}}$$

  which is equivalent to

$$C = C_j P_j^{\varepsilon} \left[\left(\int_0^1 P_i^{1-\varepsilon} di\right)^{\frac{1}{1-\varepsilon}}\right]^{-\varepsilon}$$

- One recognize the expression of the price level P, such that one gets

$$C_j = \left(\frac{P_j}{P}\right)^{-\varepsilon} C$$

5.4.2 Firms

- Each firm i is a monopoly, that faces demand function (e), taking C_t and P_t as given (monopolistic competition)

$$Y(i)_t = A_t N_t(i)^{1-\alpha}$$

- The modeling is the one of CALVO (1983)
- Each period, a firm has a probability $1-\theta$ of being allowed to reset its price
- In the aggregate, a fraction $1-\theta$ of firms resets price, a fraction θ does not.
- Duration of a price :

length in period	probability
1	$(1-\theta)$
2	$\theta(1-\theta)$
3	$\theta^2(1-\theta)$
...	...

- Average duration is therefore :

$$
\begin{aligned}
& 1\times(1-\theta)+2\times\theta(1-\theta)+3\times\theta^2(1-\theta)+4\times\theta^3(1-\theta)+\cdots \\
= \;& (1+2\theta+3\theta^2+4\theta^3+\cdots)\times(1-\theta) \\
= \;& (1+\theta+\theta^2+\theta^3+\cdots \\
& +\theta+\theta^2+\theta^3+\cdots \\
& +\theta^2+\theta^3+\cdots \\
& +\theta^3+\cdots \\
& +\cdots)\times(1-\theta) \\
= \;& \left(\tfrac{1}{1-\theta}+\tfrac{\theta}{1-\theta}+\tfrac{\theta^2}{1-\theta}+\cdots\right)\times(1-\theta) \\
= \;& 1+\theta+\theta^2+\theta^3+\cdots \\
= \;& \tfrac{1}{1-\theta}
\end{aligned}
$$

- θ is therefore an index of price stickiness.

- A firm reoptimizing in period t chooses $P_t^\star$ in order to

$$\max_{P_t^\star} \quad \sum_{j=0}^{\infty} \theta^j E_t \left[Q_{t,t+j} \left(P_t^\star Y_{t+j|t} - \Psi_{t+j}(Y_{t+j|t}) \right) \right]$$

$$\text{s.t.} \qquad Y_{t+j|t} = \left(\frac{P_t^\star}{P_{t+j}} \right)^{-\varepsilon} C_{t+j}$$

- $Y_{t+j|t}$ is the demand addressed in $t+j$ to a firm that set its price in t, and $\Psi_{t+j}(\cdot)$ is the cost function is $t+j$

5.4.2 Firms

Optimal price setting

- Consider a period j profit maximization problem :

$$\begin{array}{ll} \max_{P_t^\star} & P_t^\star Y_{t+j|t} - \Psi_{t+j}(Y_{t+j|t}) \\ \text{s.t.} & Y_{t+j|t} = \left(\frac{P_t^\star}{P_{t+j}}\right)^{-\varepsilon} C_{t+j} \end{array}$$

- The FOC is

$$Y_{t+j|t} + P_t^\star \times (-\varepsilon)(P_t^\star)^{-\varepsilon-1}\left(\frac{1}{P_{t+j}}\right)^{-\varepsilon} C_{t+j} - \underbrace{\Psi'}_{\psi} \times$$

$$(-\varepsilon)(P_t^\star)^{-\varepsilon-1}\left(\frac{1}{P_{t+j}}\right)^{-\varepsilon} C_{t+j} = 0$$

$$\iff Y_{t+j|t}\left(P_t^\star - \underbrace{\frac{\varepsilon}{1-\varepsilon}}_{\mathcal{M}}\psi_{t+j|t}\right) = 0$$

- The FOC of the intertemporal problem is therefore :

$$\sum_{j=0}^{\infty} \theta^j E_t \left[Q_{t,t+j} Y_{t+j|t} \left(P_t^{\star} - \mathcal{M} \psi_{t+j|t} \right) \right] = 0$$

- If $\theta = 0$, then $P_t^\star = \mathcal{M}\psi_t$.

Optimal price setting–Remark 2

- The choice of $P_t^\star$ is purely forward-looking. All firms resetting price will choose the same $P_t^\star$
- Rewrite the FOC in real terms, divide by P_{t-1}, and use $\Pi_{t+j,t} = \frac{P_{t+j}}{P_t}$:

$$\sum_{j=0}^{\infty} \theta^j E_t \left[Q_{t,t+j} Y_{t+j|t} \left(\frac{P_t^\star}{P_{t-1}} - \mathcal{M} \underbrace{MC_{t+j,t}}_{\text{real marginal cost}} \Pi_{t+j,t-1} \right) \right] = 0 \qquad (d)$$

- Consider again a zero inflation SS. At the SS, we have

$$\frac{P_t^\star}{P_{t-1}} = 1 \quad , \quad \Pi_{t-1,t+j} = 1 \quad , \quad P_t^\star = P_{t+j} \quad \forall \;\; t, j$$

- and therefore $Y_{t+j,t} = Y$, $MC_{t+j,t} = MC$, $Q_{t,t+j} = \beta^j$, $P^\star = \mathcal{M}\psi = \mathcal{M} \times MC \times P$, $MC = \frac{1}{\mathcal{M}}$.

5.4.2 Firms

Optimal price setting–Remark 2

- Take a first order expansion of (d) around the zero inflation SS : (Step 1)

$$p_t^\star - p_{t-1} = (1-\beta\theta)\sum_{j=0}^{\infty}(\beta\theta)^j E_t\left[\widehat{mc}_{t+j|t} + p_{t+j} - p_{t-1}\right]$$

with $\widehat{mc}_{t+j|t} = mc_{t+j|t} - mc$, $mc = -\mu$, $\mu = \log\mathcal{M}$.

- This equation can also be written

$$p_t^\star = \mu + (1-\beta\theta)\sum_{j=0}^{\infty}(\beta\theta)^j E_t\left[mc_{t+j|t} + p_{t+j}\right]$$

- The firm chooses a desired markup over a weighted average of current and expected nominal mc.

5.4.2 Firms

Aggregate Price Dynamics

- Let $s(t) \subset [0,1]$ be the set of firms that do not reset their price in period t.
- As seen previously, $P_t^\star$ is the same for all resetting firms, so that

$$P_t = \left[\int_{s(t)} P_{t-1}^{1-\varepsilon}(i)di + (1-\theta)\left(P_t^\star\right)^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}}$$

$$\iff \Pi_t^{1-\varepsilon} = \left[\theta + (1-\theta)\left(\frac{P_t^\star}{P_{t-1}}\right)^{1-\varepsilon} \right]$$

- log-linearizing around the non stochastic zero inflation SS (skipping some algebra) (Step 2) :

$$\pi_t = (1-\theta)(p_t^\star - p_{t-1})$$

5.4.3. Equilibrium

- The key difference with the "classical" model is that money matters for real equilibrium allocations

- We have

$$Y_t(i) = C_t(i) \;\; \forall \;\; i \;\; \forall \;\; t$$

- which gives by aggregation

$$Y_t = C_t$$

- and replacing in the Hh Euler equation :

$$y_t = E_t y_{t+1} - \frac{1}{\sigma}(i_t - E_t \pi_{t+1} - \rho)$$

5.4.3. Equilibrium

Labor Market

- We have

$$N(t)=\int_0^1 N_t(i)di=\int_0^1\left(\frac{Y_t(i)}{A_t}\right)^{\frac{1}{1-\alpha}}di=\left(\frac{Y_t}{A_t}\right)^{\frac{1}{1-\alpha}}\int_0^1\left(\frac{P_t(i)}{P_t}\right)^{\frac{-\varepsilon}{1-\alpha}}di$$

- which gives in logs

$$(1-\alpha)n_t=y_t-a_t+d_t$$

- where

$$d_t=(1-\alpha)\log\left[\int_0^1\left(\frac{P_t(i)}{P_t}\right)^{\frac{-\varepsilon}{1-\alpha}}di\right]$$

- One can show (with some algebra) that around a zero inflation SS, $d_t \approx 0+\mathcal{O}(d_t^2)$ so that, to a first order, (Step 3)

$$(1-\alpha)n_t=y_t-a_t$$

5.4.3. Equilibrium

The New-Keynesian Philips Curve

- Let mc_t be the (log) average marginal cost and mpn_t be the log (average) marginal product of labor :

$$
\begin{aligned}
mc_t &= (w_t - p_t) \quad -mpn_t \\
&= (w_t - p_t) \quad -(a_t - \alpha n_t + \log(1-\alpha)) \\
&= (w_t - p_t) \quad -\tfrac{1}{1-\alpha}(a_t - \alpha y_t) - \log(1-\alpha)
\end{aligned}
$$

- For a firm that does not reoptimize between t and $t+j$:

$$
\begin{aligned}
mc_{t+j|t} &= (w_{t+j} - p_{t+j}) - \tfrac{1}{1-\alpha}(a_{t+j} - \alpha y_{t+j|t}) - \log(1-\alpha) \\
&= mc_{t+j} + \tfrac{\alpha}{1-\alpha}(y_{t+j|t} - y_{t+j}) \\
&= mc_{t+j} - \tfrac{\alpha\varepsilon}{1-\alpha}(p_t^\star - p_{t+j})
\end{aligned}
$$

(using firm demand)

- We have shown that the pricing decision was *forward-looking* :

$$p_t^\star - p_{t-1} = (1-\beta\theta)\sum_{j=0}^{\infty}(\beta\theta)^j E_t[\underbrace{\widehat{mc}_{t+j|t}}_{\widehat{mc}_{t+j}+\frac{\alpha\varepsilon}{1-\alpha}(p_t^\star - p_{t+j})} + p_{t+j} - p_{t-1}]$$

- Rearranging terms, we obtain (Step 4)

$$\pi_t = \beta E_t \pi_{t+1} + \lambda \widehat{mc}_t$$

with $\lambda = \frac{(1-\theta)(1-\beta\theta)}{\theta}\Theta$ and $\Theta = \frac{1-\alpha}{1-\alpha+\alpha\varepsilon}$

- note that

$$\lambda = \lambda(\underset{-}{\theta}, \underset{-}{\alpha}, \underset{-}{\varepsilon})$$

Assignment 3

Take slides 64 to 71. Write down all the algebra needed to go through those 8 slides. In particular, make sure that you have made explicit STEPS 1 TO 4 that are indicated in the slides.

- Solving forward :

$$\pi_t = \lambda \sum_{j=0}^{\infty} \beta E_t \widehat{mc}_{t+j}$$

where $\widehat{mc}_{t+j}$ is the average mc and is also $-\mu_t$ (minus the average markup).

- When average markups are expected to be below SS, *mc* are expected to be high and inflation is high.
- To obtain a Philips Curve-like equation, let's substitute *mc* for output :

$$mc_t = (w_t - p_t) - mpn_t = \underbrace{(\sigma y_t + \phi n_t)}_{\text{labour demand}} - \underbrace{(y_t - n_t + \log(1-\alpha))}_{\text{labor supply}}$$

$$mc_t = \left(\sigma + \frac{\phi + \alpha}{1-\alpha}\right) y_t - \frac{1+\phi}{1-\alpha} a_t - \log(1-\alpha)$$

- Under flex-price, $mc_t = -\mu$, so that we can recover from the above equation the flex-price level of ouptut, denoted y^n (the *natural level of output*) :

$$y_t^n = \psi_{ya}^n a_t + \theta_y^n$$

and

$$\widehat{mc}_t = mc_t - (-\mu) = \sigma + \left(\frac{\phi + \alpha}{1 - \alpha}\right) \underbrace{(y_t - y_t^n)}_{\text{output gap } \widehat{y}_t}$$

- Therefore

$$\pi_t = \beta E_t \pi_{t+1} + \kappa \widehat{y}_t$$

- Take the Hh Euler equation and introduce the output gap.

$$y_t = E_t y_{t+1} - \frac{1}{\sigma}(i_t - E_t \pi_{t+1} - \rho)$$

- Define the *natural interest rate* as the flex-price one :

$$r_t^n = \rho + \sigma E_t \Delta y_{t+1}^n = \rho + \sigma \psi_{ya}^n E_t \Delta a_{t+1}$$

- We obtain

$$\widehat{y}_t = E_t \widehat{y}_{t+1} - \frac{1}{\sigma}(i_t - E_t \pi_{t+1} - r_t^n)$$

- One can see the model as having a recursive structure :
- natural real interest rate :
  $r_t^n = \rho + \sigma E_t \Delta y_{t+1}^n = \rho + \sigma \psi_{ya}^n E_t \Delta a_{t+1}$
- The NKPC determines π_t for a given path of $\widehat{y}_t$:
  $\pi_t = \beta E_t \pi_{t+1} + \kappa \widehat{y}_t$
- The DIS determines $\widehat{y}_t$ for a natural and actual real interest rate : $\widehat{y}_t = E_t \widehat{y}_{t+1} - \frac{1}{\sigma}(i_t - E_t \pi_{t+1} - r_t^n)$
- A monetary policy equation is still needed, which will be non neutral.

- Assume a TAYLOR rule

$$i_t = \underbrace{\rho}_{\substack{\text{consistent}\\ \text{with zero SS}\\ \text{inflation}}} + \phi_\pi \pi_t + \phi_y \widetilde{y}_t + v_t$$

  with $\phi_\pi > 0$ and $\phi_y > 0$.

- The full model is given by :

$$\begin{cases} r_t^n &= \rho + \sigma \psi_{ya}^n E_t \Delta a_{t+1} \\ \widetilde{y}_t &= E_t \widetilde{y}_{t+1} - \frac{1}{\sigma}(i_t - E_t \pi_{t+1} - r_t^n) \\ \pi_t &= \beta E_t \pi_{t+1} + \kappa \widetilde{y}_t \\ a_t &= \rho_a a_{t-1} + \varepsilon_t^a \\ v_t &= \rho_v v_{t-1} + \varepsilon_t^v \end{cases}$$

- or equivalently

$$\underbrace{\begin{pmatrix} \widetilde{y}_t \\ \pi_t \end{pmatrix}}_{X_t} = \underbrace{\Omega}_{\frac{1}{\sigma+\phi_y+\kappa\phi_\pi}} \underbrace{\begin{bmatrix} \sigma & 1-\beta\phi_\pi \\ \sigma\kappa & \kappa+\beta(\sigma+\phi_t) \end{bmatrix}}_{A} \times \begin{pmatrix} E_t\widetilde{y}_{t+1} \\ E_t\pi_{t+1} \end{pmatrix} + \Omega \underbrace{\begin{pmatrix} 1 \\ \kappa \end{pmatrix}}_{B} \underbrace{(\widehat{r}_t - v_t)}_{z_t}$$

- This equation can be solved forward to obtain a unique solution if and only if eig$(A)< 1$, which is guaranteed if $\kappa(\phi_\pi - 1) + (1-\beta)\phi_y > 0$
- and the solution is

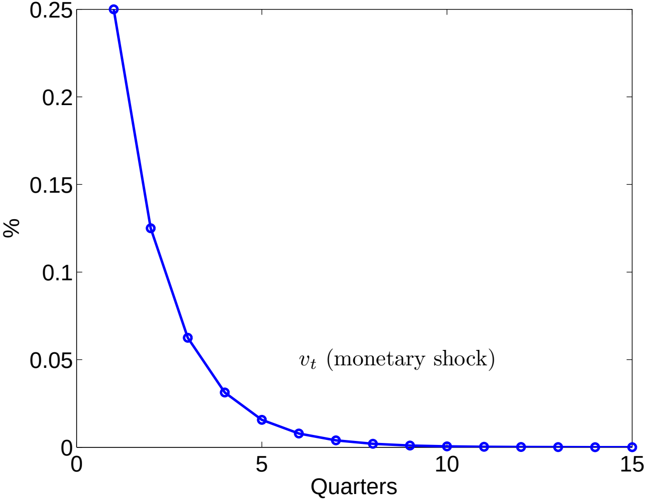
$$X_t = \sum_{j=0}^{\infty} A^j E_t z_{t+j}$$

Calibration

β	.99	σ	1
ϕ	1	α	1/3
ε	6	η	4
θ	2/3	ϕ_π	1.5
ϕ_y	.5/4	ρ_a	.9
ρ_v	.5		

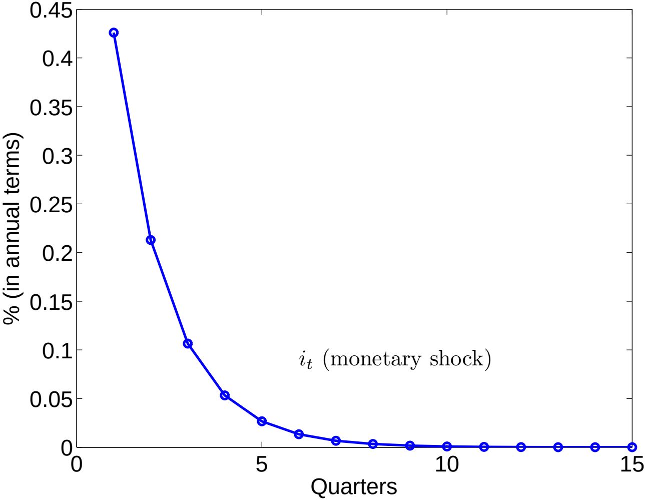
5.4.4. Equilibrium under an Interest Rate Rule

Monetary Shock



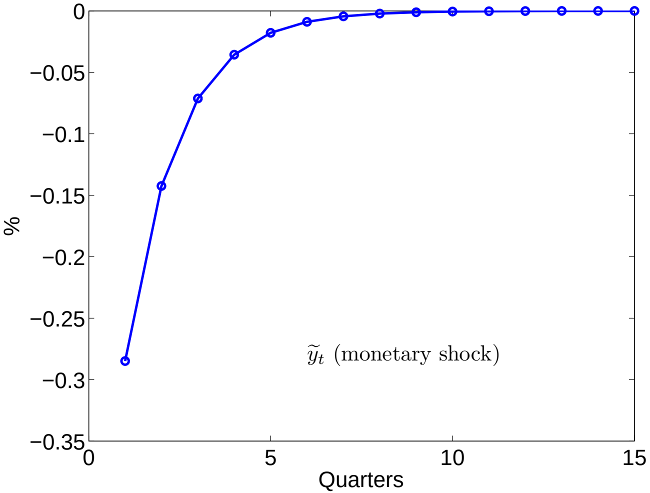
5.4.4. Equilibrium under an Interest Rate Rule

Monetary Shock



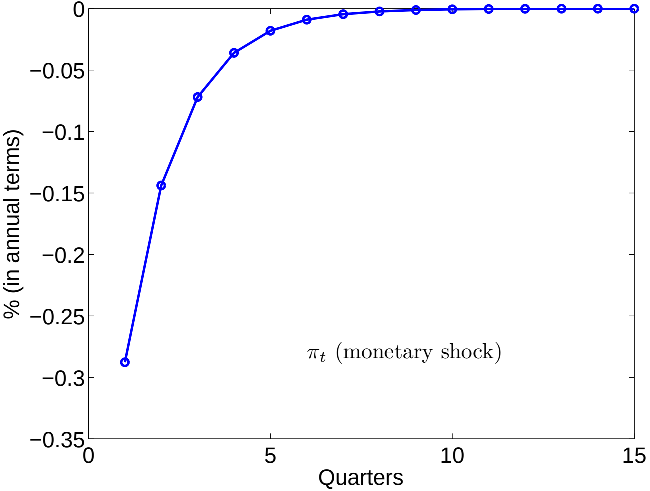
5.4.4. Equilibrium under an Interest Rate Rule

Monetary Shock



5.4.4. Equilibrium under an Interest Rate Rule

Monetary Shock



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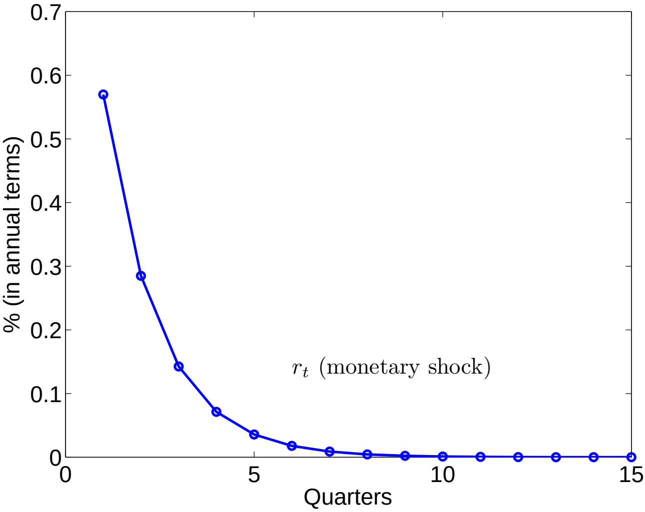
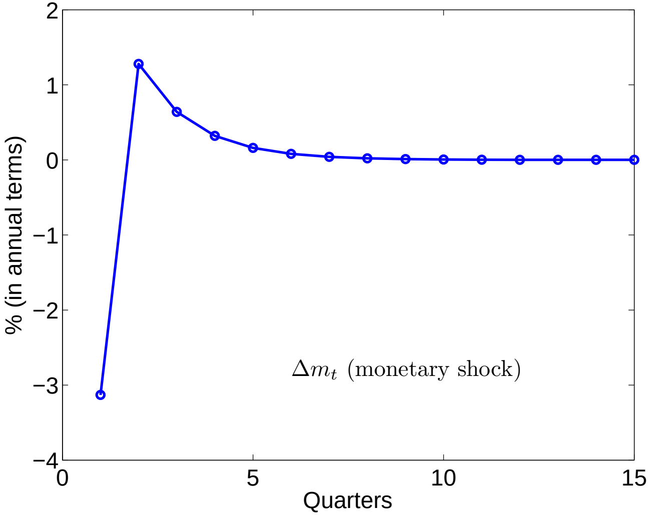
Monetary Shock



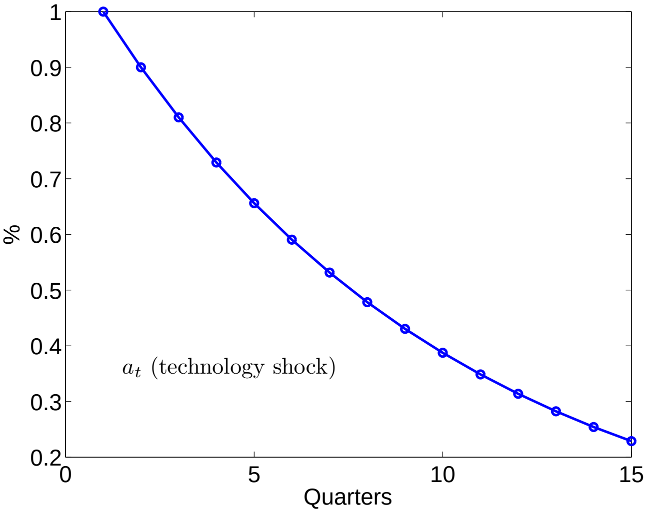
5.4.4. Equilibrium under an Interest Rate Rule

Monetary Shock



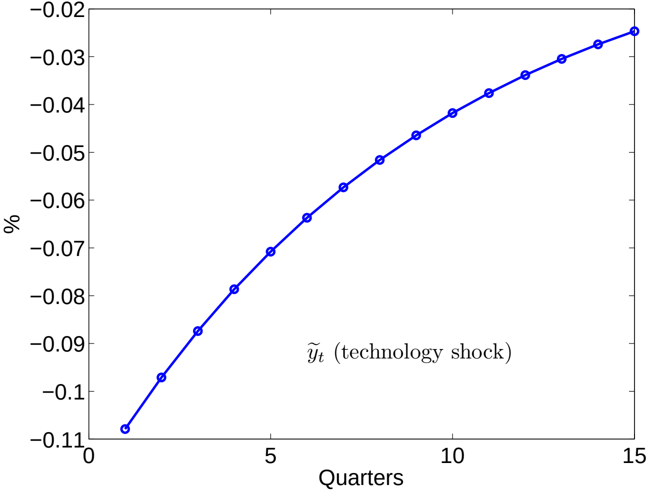
5.4.4. Equilibrium under an Interest Rate Rule

Technology Shock



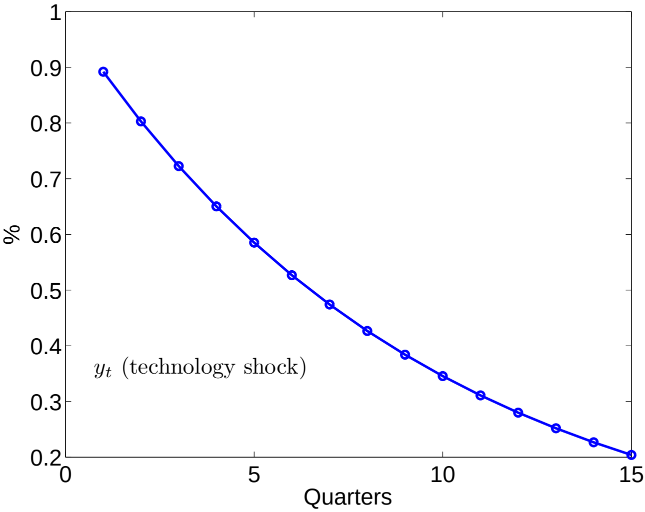
5.4.4. Equilibrium under an Interest Rate Rule

Technology Shock



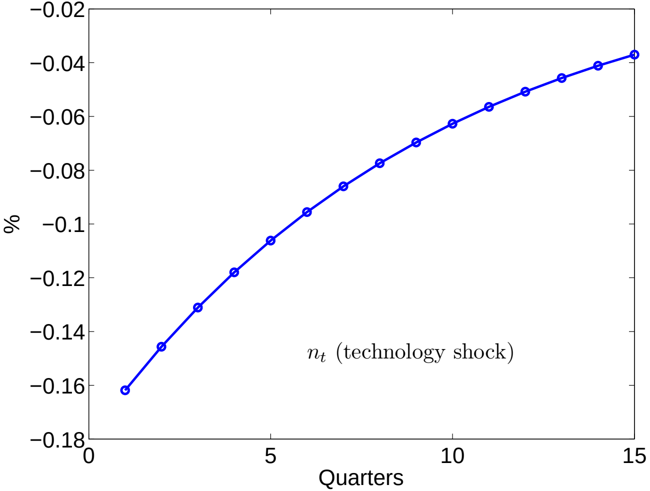
5.4.4. Equilibrium under an Interest Rate Rule

Technology Shock



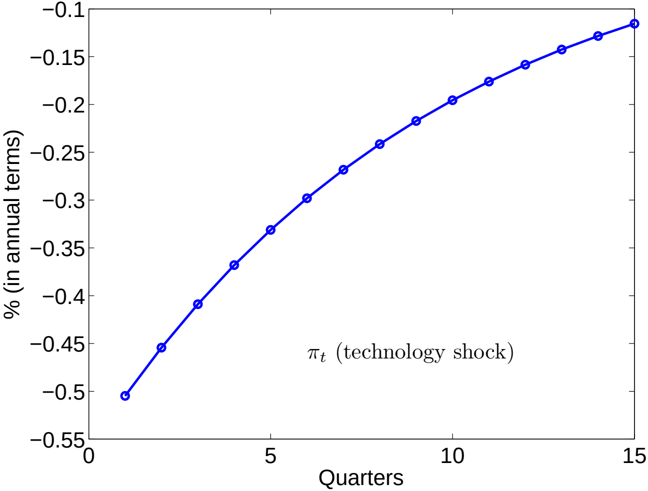
5.4.4. Equilibrium under an Interest Rate Rule

Technology Shock



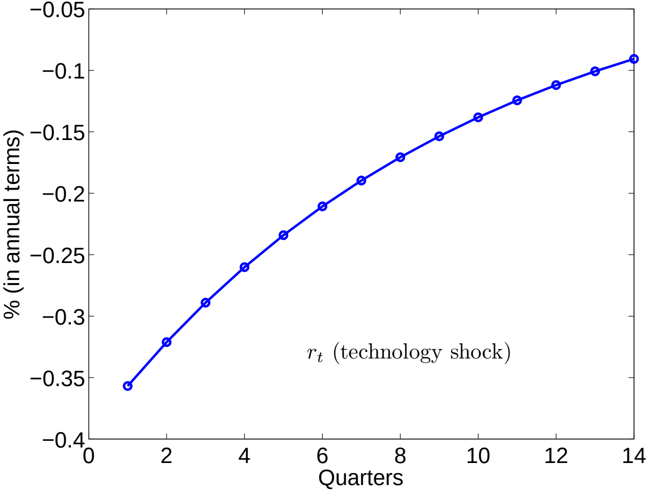
5.4.4. Equilibrium under an Interest Rate Rule

Technology Shock



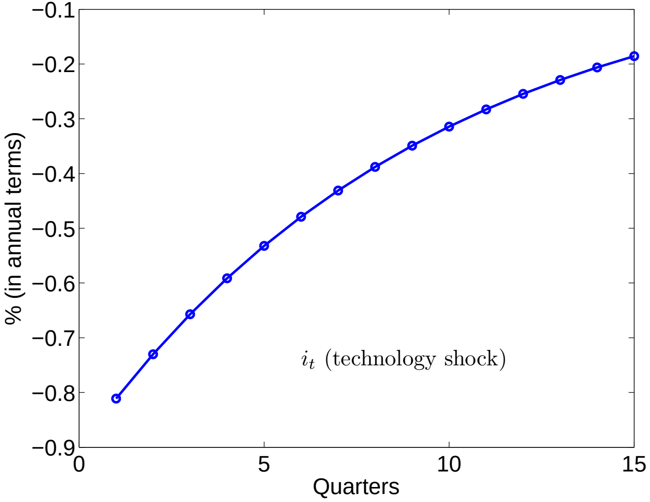
5.4.4. Equilibrium under an Interest Rate Rule

Technology Shock



5.4.4. Equilibrium under an Interest Rate Rule

Technology Shock



5.4.4. Equilibrium under an Interest Rate Rule

Technology Shock